

Name _____

ID number _____ Date _____ Group _____

“Adhering to the Code of Ethics for students of the Tec de Monterrey, I pledge to follow the precepts of academic honesty in this evaluation by not using unauthorized or illicit resources to solve it.”

Signature

I. Read the following questions and identify the option that best answers each, then write its letter on the line. (5 points each)

1. _____ Given the function $f(x) = -2\sqrt{x^2 + 4} - 5$, which of the following represents its domain?
a) $(-\infty, \infty)$ b) $[-4, \infty)$ c) $(-\infty, -4) \cup (-4, \infty)$ d) $(-4, \infty)$

2. _____ A small publishing company is releasing a new book. The production costs is $C(N) = 10\sqrt{N} + 150$ and each book is going to be sell at 2 dollars each $P(N) = 2N$. Which of the following equations will represent $(C - P)(N)$?

- a) $(C - P)(N) = -10\sqrt{N} + 150 + 2N$ b) $(C - P)(N) = 2(5\sqrt{N} + 75 - N)$
c) $(C - P)(N) = -2N - 10\sqrt{N} + 150$ d) $(C - P)(N) = 2N - 10\sqrt{N} - 150$

3. _____ Let $g(x) = \frac{4x}{x-1}$ and $f(x) = \sqrt{x} - 5$, then perform $g(2)$. Which of the following options represents the result?

- a) 7 b) 8 c) 4 d) $\frac{14}{3}$

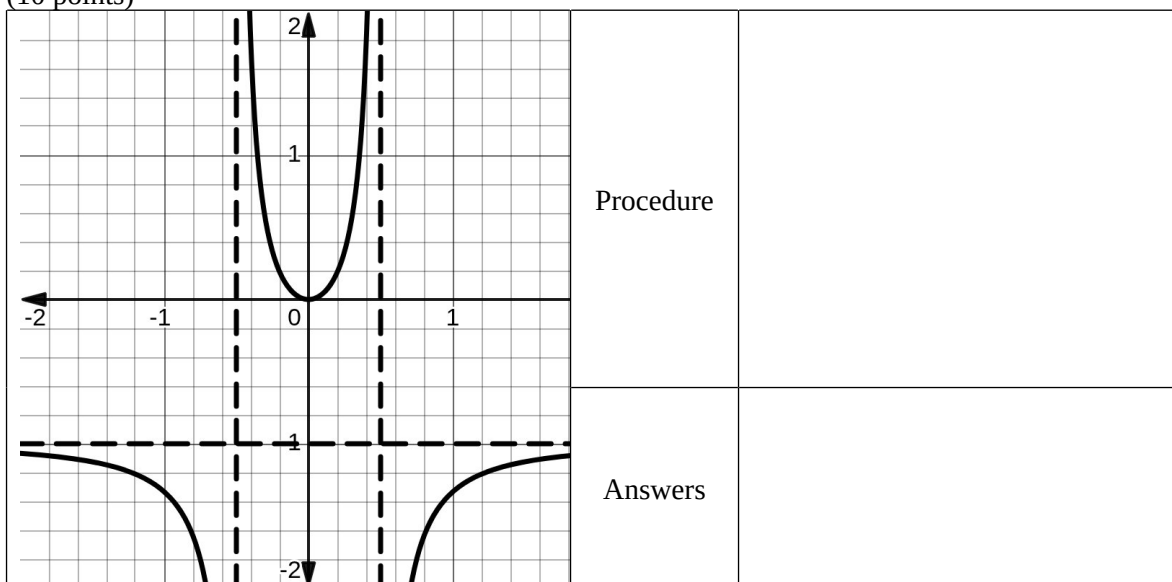
4. _____ The following piecewise function represents the cost of bathing a dog in the new grooming saloon. The cost depends on the weight of the dog, in kilograms. Which would be the cost for a dog that weighs 5 kg ?

$$C(w) = \begin{cases} 150, & w < 5 \text{ kg} \\ 150 + 5w, & 5 \leq w < 12 \\ 175 + 50w, & w \geq 12 \end{cases}$$

- a) \$234 b) \$765 c) \$175 d) \$150

II. Answer the following exercises.

1. Determine the domain and range of the rational function depicted in the following graph. (10 points)



2. Determine the vertical and horizontal asymptotes of the following equation. (10 points)

$$f(x) = \frac{2x - 1}{x^2 - x - 2}$$

Procedure	
Answer	

3. Determine the intersections with x and y. Write the final answers as coordinates. (10 points)

$$f(x) = \frac{x^2 + x - 6}{x^2 - 25}$$

Procedure	
Answer	

III. Solve the following exercises in an orderly and clear manner. **Answers without procedure will not be considered.** Underline or frame your final answer.

4. Determine whether f and g are inverses of each other or not. Simplify your answer as much as possible. Justify your answer using $f(g(x))$ and $g(f(x))$. (10 points)

$$f(x) = \frac{x-1}{2} \text{ and } g(x) = 2x + 1$$

Procedure	
Answer	

5. Find $h^{-1}(x)$, given the following function $h(x) = (x + 5)^2$ for the domain $[-5, \infty)$. (10 points)

Procedure	
Answer	

6. Find the inverse function of $f(x) = \frac{5x-4}{x+3}$ (10 points)

Procedure	
Answer	

7. Define the coordinate of the empty hole. (10 points)

$$f(x) = \frac{x^2}{0.5(x-2)3x}$$

Procedure	
Answer	

5. Graph the piecewise function. (10 points)

$$f(x) = \begin{cases} x^2, & x \in [0, 2] \\ \frac{(x-10)^3}{48} + \frac{44}{3}, & x \in [2, 30] \\ 50\sqrt{x-29} + \frac{394}{3}, & x \in [30, 35] \end{cases}$$

